

The RAG Illusion: Why Retrieval Accuracy Doesn't Guarantee Faithfulness

Your enterprise AI might be retrieving the correct documents, but it is answering the wrong questions. Discover the architectural fix for hallucinated structures.

Executive Summary:

In enterprise Retrieval-Augmented Generation (RAG) systems, retrieving the correct topical context is only half the battle. A common, silent failure occurs when an LLM is asked a multi-part question. The system retrieves topically relevant passages, but the LLM maps the structure of the question onto loosely related facts, creating a highly confident, entirely misattributed answer. This whitepaper details the implementation of Query Decomposition and Cross-Encoder Re-ranking to enforce absolute structural faithfulness in production AI.

1. The Engineering Challenge

During the engineering of a highly specialized bilingual RAG system, we encountered a critical failure mode: 'faithfulness failure under weak grounding.' When asked a compound question (e.g., 'What is X, and what are its types?'), the embedding search retrieved chunks about X. However, the chunks described locations, not types. The LLM, eager to fulfill the 'types' prompt, hallucinated a list by mapping the locations into the 'types' structure. Standard system prompts ('answer only from context') failed to prevent this because the model interpreted the loosely related content as the answer.

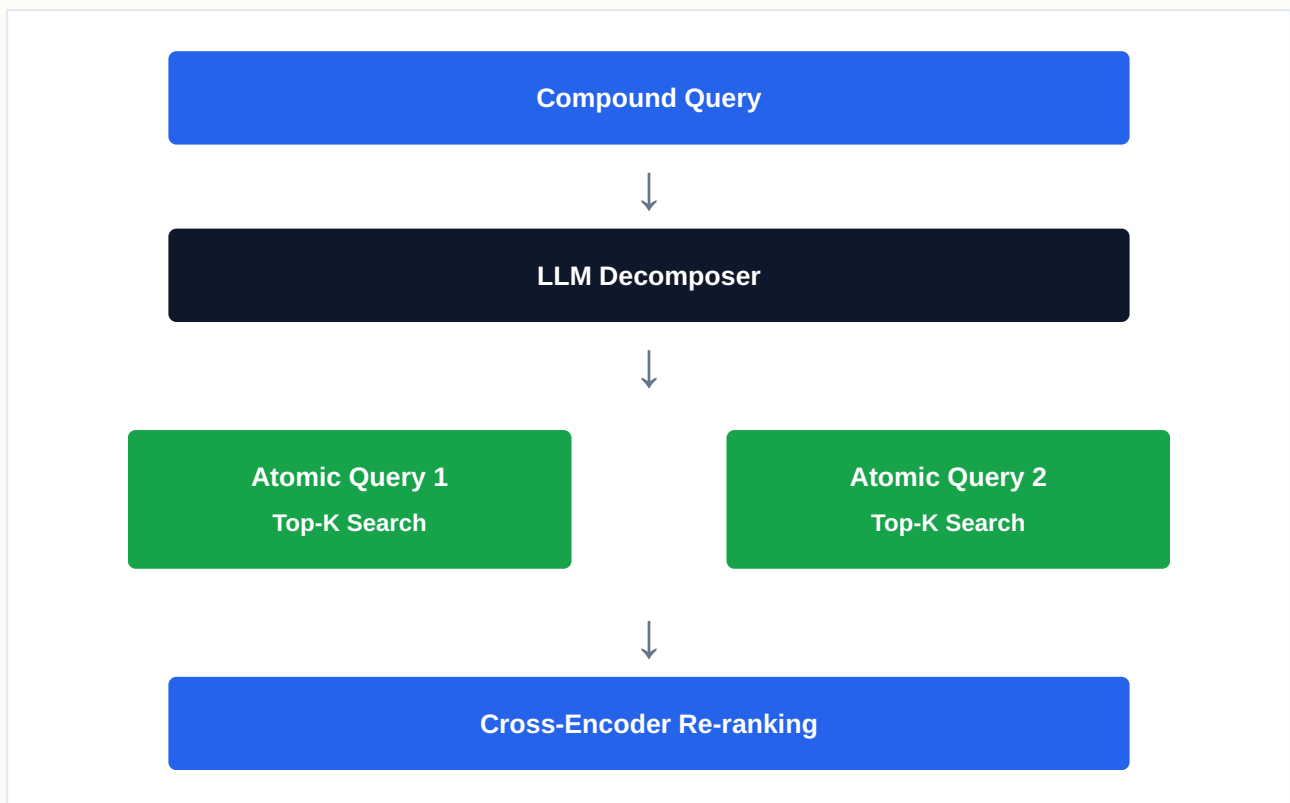
In high-stakes enterprise environments, this kind of failure is unacceptable. When building systems for Fortune 500 organizations, relying on basic prompt engineering or out-of-the-box vendor SDKs frequently masks underlying architectural fragility. The goal is to move beyond simple proofs-of-concept into deterministic, resilient data flow.

2. The Architectural Solution

We architected a multi-stage retrieval defense. First, compound queries are intercepted by an LLM-driven Decomposer that breaks the prompt into atomic sub-queries. Second, each sub-query executes an independent vector search. Finally, all retrieved chunks are merged, deduplicated, and passed through a Cross-Encoder Re-ranker (e.g., ms-marco-MiniLM) to score each chunk against the specific atomic question, eliminating topically adjacent but factually useless data.

By enforcing this pattern at the architectural level, the system no longer relies on the "magic" of generative fluency. Instead, it relies on rigorous data engineering and precise, targeted models. This decoupling ensures that failures are isolated, auditable, and easily corrected without unwinding the entire data pipeline.

3. Structural Data Flow



4. Business Impact & ROI

For organizations operating at scale, migrating to this pattern fundamentally alters the cost and reliability curve. It shifts the burden of trust from the LLM's inherently probabilistic

nature to a deterministic data engineering layer. The result is a system that can process massive throughput securely, drastically reducing API token burn and ensuring complete auditability for internal compliance boards.